

TASK 1:

```
MySQL 8.0 Command Line Cli  ×  +  v

Query OK, 1 row affected (0.01 sec)

mysql> USE `2520090081`;
Database changed
mysql> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| 2520090081 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM student_marks;
ERROR 1146 (42S02): Table '2520090081.student_marks' doesn't exist
mysql> USE `2520090081`;
Database changed
mysql> CREATE TABLE student_marks (
->     roll_no INT PRIMARY KEY,
->     name VARCHAR(50),
->     subject VARCHAR(50),
->     marks DECIMAL(5,2)
-> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO student_marks (roll_no, name, subject, marks) VALUES
-> (1, 'Ravi', 'Math', 85.50),
-> (2, 'Sita', 'Math', 92.75),
-> (3, 'Anil', 'Math', 78.40),
-> (4, 'Priya', 'Math', 88.90),
-> (5, 'Vijay', 'Math', 80.25),
-> (6, 'subbusir', 'aws', 98.50),
-> (7, 'dwaraka', 'dbms', 95.75),
-> (8, 'ranjani', 'english', 97.40),
-> (9, 'kavithamam', 'aws1', 99.90),
-> (10, 'seetha', 'azure', 82.25);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> |
```

Task 2: Insert Student Data

```
MySQL 8.0 Command Line Cli  ×  +  ▾

mysql> INSERT INTO student_marks (roll_no, name, subject, marks) VALUES
-> (1, 'Ravi', 'Math', 85.50),
-> (2, 'Sita', 'Math', 92.75),
-> (3, 'Anil', 'Math', 78.40),
-> (4, 'Priya', 'Math', 88.90),
-> (5, 'Vijay', 'Math', 80.25),
-> (6, 'subbusir', 'aws', 98.50),
-> (7, 'dwaraka', 'dbms', 95.75),
-> (8, 'ranjani', 'english', 97.40),
-> (9, 'kavithamam', 'aws1', 99.90),
-> (10, 'seetha', 'azure', 82.25);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM student_marks;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi      | Math    | 85.50 |
| 2       | Sita      | Math    | 92.75 |
| 3       | Anil      | Math    | 78.40 |
| 4       | Priya     | Math    | 88.90 |
| 5       | Vijay     | Math    | 80.25 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha    | azure   | 82.25 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> |
```

Task 3: Retrieve Student Data

```
MySQL 8.0 Command Line Cli  X  +  v
ERROR 1062 (23000): Duplicate entry '1' for key 'student_marks.PRIMARY'
mysql> USE `2520090081`;
Database changed
mysql> SELECT * FROM student_marks;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi   | Math    | 85.50 |
| 2       | Sita   | Math    | 92.75 |
| 3       | Anil   | Math    | 78.40 |
| 4       | Priya  | Math    | 88.90 |
| 5       | Vijay  | Math    | 80.25 |
| 6       | subbusir | aws    | 98.50 |
| 7       | dwaraka | dbms    | 95.75 |
| 8       | ranjani | english | 97.40 |
| 9       | kavithamam | aws1   | 99.90 |
| 10      | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> USE `2520090081`;
Database changed
mysql> SELECT * FROM student_marks;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi   | Math    | 85.50 |
| 2       | Sita   | Math    | 92.75 |
| 3       | Anil   | Math    | 78.40 |
| 4       | Priya  | Math    | 88.90 |
| 5       | Vijay  | Math    | 80.25 |
| 6       | subbusir | aws    | 98.50 |
| 7       | dwaraka | dbms    | 95.75 |
| 8       | ranjani | english | 97.40 |
| 9       | kavithamam | aws1   | 99.90 |
| 10      | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> |
```

TASK 4: Aggregate Functions

```
MySQL 8.0 Command Line Cli  X  +  v

mysql> INSERT INTO student_marks (roll_no, name, subject, marks) VALUES
-> (1, 'Ravi', 'Math', 85.50),
-> (2, 'Sita', 'Math', 92.75),
-> (3, 'Anil', 'Math', 78.40),
-> (4, 'Priya', 'Math', 88.90),
-> (5, 'Vijay', 'Math', 80.25),
-> (6, 'subbusir', 'aws', 98.50),
-> (7, 'dwaraka', 'dbms', 95.75),
-> (8, 'ranjani', 'english', 97.40),
-> (9, 'kavithamam', 'aws1', 99.90),
-> (10, 'seetha', 'azure', 82.25);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM student_marks;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi      | Math    | 85.50 |
| 2       | Sita      | Math    | 92.75 |
| 3       | Anil      | Math    | 78.40 |
| 4       | Priya     | Math    | 88.90 |
| 5       | Vijay     | Math    | 80.25 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha    | azure   | 82.25 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> |
```

Task 6: Advanced Filtering

```
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mysql>
mysql>
mysql>
mysql> SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi  | Math    | 85.50 |
| 4       | Priya | Math    | 88.90 |
| 5       | Vijay | Math    | 80.25 |
| 10      | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM student_marks WHERE name LIKE 'P%';
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
| 4       | Priya | Math    | 88.90 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT * FROM student_marks
    -> WHERE name IN ('Ravi', 'Sita', 'Vijay');
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi  | Math    | 85.50 |
| 2       | Sita  | Math    | 92.75 |
| 5       | Vijay | Math    | 80.25 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM student_marks
    -> WHERE marks > 85
    -> AND (subject = 'Math' OR name LIKE 'P%');|
```

Task 7: Update Student Records

```
MySQL 8.0 Command Line Cli  X + v

mysql>
mysql>
mysql>
mysql>
mysql>
mysql>
mysql>
mysql> UPDATE student_marks
    -> SET marks = 90.00
    -> WHERE roll_no = 3;
Query OK, 1 row affected (0.08 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE student_marks
    -> SET subject = 'Remedial Math'
    -> WHERE marks < 80;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> SELECT * FROM student_marks;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi      | Math    | 85.50 |
| 2       | Sita      | Math    | 92.75 |
| 3       | Anil      | Math    | 90.00 |
| 4       | Priya     | Math    | 88.90 |
| 5       | Vijay     | Math    | 80.25 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha    | azure   | 82.25 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> |
```

Task 8: Delete Student Records

```
MySQL 8.0 Command Line Cli × + v
```

8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

10 rows in set (0.00 sec)

```
mysql>
mysql>
mysql>
mysql>
mysql>
mysql>
mysql> DELETE FROM student_marks
    -> WHERE roll_no = 5;
Query OK, 1 row affected (0.03 sec)

mysql>
mysql> DELETE FROM student_marks
    -> WHERE marks < 75;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT * FROM student_marks;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

9 rows in set (0.00 sec)

```
mysql> |
```

Task 9: Sorting Records

```
MySQL 8.0 Command Line Cli  ×  +  v

9 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks ORDER BY marks ASC;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
| 10      | seetha | azure   | 82.25 |
| 1       | Ravi   | Math    | 85.50 |
| 4       | Priya  | Math    | 88.90 |
| 3       | Anil   | Math    | 90.00 |
| 2       | Sita   | Math    | 92.75 |
| 7       | dwaraka | dbms    | 95.75 |
| 8       | ranjani | english | 97.40 |
| 6       | subbusir | aws     | 98.50 |
| 9       | kavithamam | aws1    | 99.90 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM student_marks ORDER BY marks DESC;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
| 9       | kavithamam | aws1    | 99.90 |
| 6       | subbusir   | aws     | 98.50 |
| 8       | ranjani    | english | 97.40 |
| 7       | dwaraka    | dbms    | 95.75 |
| 2       | Sita       | Math    | 92.75 |
| 3       | Anil       | Math    | 90.00 |
| 4       | Priya      | Math    | 88.90 |
| 1       | Ravi       | Math    | 85.50 |
| 10      | seetha     | azure   | 82.25 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM student_marks ORDER BY name ASC;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
```

9	kavithamam	aws1	99.90
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9 rows in set (0.00 sec)

mysql>

mysql> SELECT * FROM student_marks ORDER BY marks DESC;

roll_no	name	subject	marks
9	kavithamam	aws1	99.90
6	subbusir	aws	98.50
8	ranjani	english	97.40
7	dwaraka	dbms	95.75
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
1	Ravi	Math	85.50
10	seetha	azure	82.25

9 rows in set (0.00 sec)

mysql>

mysql> SELECT * FROM student_marks ORDER BY name ASC;

roll_no	name	subject	marks
3	Anil	Math	90.00
7	dwaraka	dbms	95.75
9	kavithamam	aws1	99.90
4	Priya	Math	88.90
8	ranjani	english	97.40
1	Ravi	Math	85.50
10	seetha	azure	82.25
2	Sita	Math	92.75
6	subbusir	aws	98.50

9 rows in set (0.08 sec)

mysql>

mysql> SELECT * FROM student_marks ORDER BY name DESC;

Task 10: GROUP BY

```
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mysql>
mysql>
mysql>
mysql>
mysql>
mysql> SELECT subject, SUM(marks) AS total_marks
        -> FROM student_marks
        -> GROUP BY subject;
+-----+-----+
| subject | total_marks |
+-----+-----+
| Math    | 357.15      |
| aws     | 98.50       |
| dbms    | 95.75       |
| english | 97.40       |
| aws1    | 99.90       |
| azure   | 82.25       |
+-----+-----+
6 rows in set (0.00 sec)

mysql>
mysql> SELECT subject, AVG(marks) AS avg_marks
        -> FROM student_marks
        -> GROUP BY subject;
+-----+-----+
| subject | avg_marks |
+-----+-----+
| Math    | 89.287500 |
| aws     | 98.500000 |
| dbms    | 95.750000 |
| english | 97.400000 |
| aws1    | 99.900000 |
| azure   | 82.250000 |
+-----+-----+
6 rows in set (0.00 sec)

mysql>
mysql> SELECT subject, COUNT(*) AS num_students
        -> FROM student_marks
        -> GROUP BY subject;
```

Task 11: HAVING

```
MySQL 8.0 Command Line Cli  ×  +  ∨

mysql>
mysql> SELECT subject, COUNT(*) AS num_students
      -> FROM student_marks
      -> GROUP BY subject;
+-----+-----+
| subject | num_students |
+-----+-----+
| Math    | 4            |
| aws     | 1            |
| dbms    | 1            |
| english | 1            |
| aws1    | 1            |
| azure   | 1            |
+-----+-----+
6 rows in set (0.00 sec)

mysql>
mysql>
mysql>
mysql>
mysql> SELECT subject, AVG(marks) AS avg_marks
      -> FROM student_marks
      -> GROUP BY subject
      -> HAVING AVG(marks) > 90;
+-----+-----+
| subject | avg_marks |
+-----+-----+
| aws     | 98.500000 |
| dbms    | 95.750000 |
| english | 97.400000 |
| aws1    | 99.900000 |
+-----+-----+
4 rows in set (0.00 sec)

mysql>
mysql> SELECT subject, COUNT(*) AS num_students
      -> FROM student_marks
      -> GROUP BY subject
      -> HAVING COUNT(*) > 1;
```